
Technology transfer package — Toscanini invention

invbench-p03-phe-biofouling156ae5155c73 · run r401-bench-p03-phe-biofouling · maturity
EARLY_TECHNICAL_EVALUATION · 3D class CONCEPTUAL_3D

Problem definition

- Problem Id: bench-p03-phe-biofouling
- Device: stainless steel plate heat exchanger in a district cooling loop
- Failure: biofilm and scale accumulation on plate channels reduces overall heat transfer coefficient and raises pressure drop between cleanings
- Failure Mode: fouling
- Constraint: sustain heat transfer coefficient above 2500 W/m²K for 12 months between mechanical cleanings without chlorination above 0.5 ppm residual

Failure mode being addressed

6 failure modes recorded. Primary: fouling — trigger: the operating condition stated in the problem (mechanism detail beyond the problem statement and trigger quantification NOT ESTABLISHED)

Mechanism

Toscanini investigated the problem “inv:bench-p03-phe-biofouling:156ae5155c73” and, after 2 architecture generations of challenge-and-evolution, its current invention is generation 2. The causal mechanism is Real-time distributed fiber optic sensing combined with microcapsule-based self-healing agents that release when corrosion is detected applied at the recorded intervention site. The expected measurable effect is Autonomous detection and repair of corrosion sites as they form, extending component lifespan by 300% and reducing maintenance costs by 70%. The invention's maturity is recorded honestly: this is an invented architecture supported by retrieved evidence — not a physically validate...

Causal delta

What distinguishes this generation from its parent is the explicit causal change recorded in the evolution: Real-time distributed fiber optic sensing combined with microcapsule-based self-healing agents that release when corrosion is detected — the architecture delta versus the parent generation is recorded in the run state and inspectable in the package.

Architecture

- heat source characterization — status ENGINEERING_PROPOSED
- conduction path / spreader — status ENGINEERING_PROPOSED
- reject interface — status ENGINEERING_PROPOSED
- safety limit interlock — status ENGINEERING_PROPOSED

Subsystem breakdown

- heat source characterization — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)
- conduction path / spreader — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)
- reject interface — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)

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- safety limit interlock — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)

Governing equations

- TH-002: $q = h * A * dT$ — Convective heat transfer (Newton's law of cooling)
 - TH-003: $R_{total} = \sum(L_i / (k_i * A_i))$ — Series thermal resistance
 - TH-001: $q = -k * A * dT/dx$ — Fourier conduction
- (3 total; full bindings in EQUATION_REGISTRY.json)

Parameter definitions

- CP-001 heat generation at source [q W] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-002 spreader conductivity [k_i W/(m.K)] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-003 interface contact resistance [R_{total} K/W] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-004 reject-path area [A m²] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-005 allowable temperature rise [dT K] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-006 ambient worst case [NOT_ASSIGNED UNKNOWN] = UNKNOWN (no sourced value) (UNKNOWN)

Constraints

- Stated Constraint: sustain heat transfer coefficient above 2500 W/m²K for 12 months between mechanical cleanings without chlorination above 0.5 ppm residual
- Failure Mode Of Device: fouling

Engineering assumptions

- Assumption: mechanism transfers across domain context
- Basis: MODELLED
- Test: Accelerated aging test with simultaneous chloride exposure and mechanical stress while monitoring healing agent deplet...
- Assumption: searched sources approximate the prior-art universe
- Basis: MODELLED
- Test: expand sources (see next_best_action)

Failure modes

- FM-001: fouling — severity UNKNOWN (no sourced severity basis)
- FM-DOM-001: hotspot exceeding material/safety limit — severity UNKNOWN (no sourced severity basis)
- FM-DOM-002: interface delamination raising resistance over life — severity UNKNOWN (no sourced severity basis)
- FM-DOM-003: coolant/flow path blockage (if active) — severity UNKNOWN (no sourced severity basis)
- FM-DOM-004: thermal runaway coupling to electronics — severity UNKNOWN (no sourced severity basis)
- FM-DOM-005: condensation at cold surfaces — severity UNKNOWN (no sourced severity basis)

Verification plan

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- VF-001: result NOT_TESTED — Accelerated aging test with simultaneous chloride exposure and mechanical stress while monitoring healing agent depletion and repair effectiveness
 - VF-002: result NOT_TESTED — run killer experiment: falsification_test_from_candidate (EIG/cost 0.4557178723356231)
 - VF-003: result NOT_TESTED — temperature-rise mapping at max duty
 - VF-004: result NOT_TESTED — temperature-rise mapping at maximum duty with IR or embedded sensing (measured quantity: temperature rise)
 - VF-005: result NOT_TESTED — thermal-resistance trend over accelerated cycling (measured quantity: mechanical stress / fatigue endurance)
 - VF-006: result NOT_TESTED — temperature-rise mapping at max duty
 - VF-007: result NOT_TESTED — worst-case ambient soak with staged power steps and stability monitoring (measured quantity: power / energy deposition)

Build / manufacturing pathway

- WP-01: test article first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed); measurement physical bench/clinical experiment — NOT simulation
- WP-02: test article constraint-compliance envelope article; measurement constraint-compliance characterization
- WP-03: test article article for VF-001; measurement Accelerated aging test with simultaneous chloride exposure and mechanical stress while monitoring healing agent depletio...
- WP-04: test article article for VF-002; measurement run killer experiment: falsification_test_from_candidate (EIG/cost 0.4557178723356231)
- WP-05: test article article for VF-003; measurement temperature-rise mapping at max duty
- WP-06: test article article for VF-004; measurement temperature-rise mapping at maximum duty with IR or embedded sensing (measured quantity: temperature rise)
- WP-07: test article article for VF-005; measurement thermal-resistance trend over accelerated cycling (measured quantity: mechanical stress / fatigue endurance)
- WP-08: test article article for VF-006; measurement temperature-rise mapping at max duty
- WP-09: test article article for VF-007; measurement worst-case ambient soak with staged power steps and stability monitoring (measured quantity: power / energy deposition)

Regulatory posture

problem context (canonical applicability): INDUSTRIAL_PROCESS. Context requirement [UNKNOWN]: applicable industrial codes/standards (e.g. pressure equipment, piping, electrical) UNKNOWN until determined for the installation jurisdiction — the buyer's regulatory affairs makes the determination;.... Medical-device regulation: NOT_APPLICABLE — the canonical problem context carries NO patient-contact/clinical-use signals — medical regulatory requirements are outside this problem's a.... regulatory pathway: NOT ESTABLISHED (no determination exists for this invention; the pathway is never inferred from device class alone — the buyer's first regulatory action is the determination itself)

Decisive experiment

- Selected: falsification_test_from_candidate
- Eig: 0.4557178723356231
- Eig Per Cost: 0.4557178723356231
- Definition: physical bench/clinical experiment — NOT simulation
- Hypotheses: • H_effect_holds
- H_effect_fails
- Options Ranked: • falsification_test_from_candidate

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- contradiction_attacking_experiment
 - Epistemic Note: all priors MODEL_DERIVED; no EXPERIMENTALLY_ESTIMATED inputs exist at this stage (Art. XXVIII/XXXVIII: no synthetic shortcut to physical observation)

Expected measurements

- Accelerated aging test with simultaneous chloride exposure and mechanical stress while monitoring healing agent depletion and repair effectiveness
 - run killer experiment: falsification_test_from_candidate (EIG/cost 0.4557178723356231)
 - temperature-rise mapping at max duty
 - temperature-rise mapping at maximum duty with IR or embedded sensing (measured quantity: temperature rise)
 - thermal-resistance trend over accelerated cycling (measured quantity: mechanical stress / fatigue endurance)
 - temperature-rise mapping at max duty
- (all NOT_TESTED — expected values are UNKNOWN until the experiment runs)

Acceptance criteria

Pre-registered rule (from VF-001): pre-registered pass/fail against the problem-stated threshold(s) (SOURCE_FACT, problem statement; the sourced values and their provenance are recorded on the corresponding design-input records)

Unresolved technical risks

The run records its unknowns in the package's engineering definition; each unknown is material and none is suppressed. Physical output is NOT ESTABLISHED — no measurement of this invention exists.

3D explanation

A conceptual architecture visualization (CONCEPTUAL_3D) — topology and named components only, NO engineering dimensions; engineering CAD is earned only when parameters are sourced (labeled in 3D_DESIGN_STATUS.json)

Artifact references

every layer of this package is named in PACKAGE_MANIFEST.json with per-file sha256; the machine-readable layers are ENGINEERING_TRACEABILITY, EQUATION_REGISTRY, UNKNOWN_ROADMAP, VALIDATION_ECONOMICS, MATURITY_BASIS, LOOP_STATE, PROVENANCE (top level) and the MODEL/ tree

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.